Mueller polarimetric imaging system with liquid crystals

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We present a new polarimetric imaging system based on liquid-crystal modulators, a spectrally filtered white-light source, and a CCD camera. The whole Mueller matrix image of the sample is measured in approximately 5 s in the transmission mode. The instrument design, together with an original and easy-to-operate calibration procedure, provides high accuracy over a wide spectral range (500–700 nm). This accuracy has been assessed by measurement of a linear polarizer at different orientations and a thick wedged quartz plate as an example of a partially depolarized retarder. Polarimetric images of a stained hepatic biopsy with significant fibrosis have been taken at several wavelengths. The optical properties of Picrosirius Red stained collagen (diattenuation, retardance, and polarizance) have been measured independently from each other between 500 and 700 nm. © 2004 Optical Society of America

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1. Introduction

The light polarization in a polarimeter can be modulated and analyzed by a variety of devices, such as rotating retardation plates,[1,2] rotating compensators,[3] Pockels cells,[4] photoelastic modulators,[5,6] or liquid-crystal variable retarders.[7,8] We recently demonstrated[9] a Mueller polarimeter in which the polarization state generator (PSG) and polarization state analyzer (PSA) are both composed of a linear polarizer and two liquid-crystal variable retarders. Liquid crystals are particularly well suited for polarimetric imaging, as these devices typically feature broad acceptance angles, wide clear apertures, low aberrations, and no image wander. In addition, they do not require any moving parts and are operated at low driving voltages, with switching frequencies well suited for acquisition with standard CCD detectors. With this approach we built an imaging Mueller polarimeter that is faster than those described previously.[10,11] In addition we easily obtained high accuracy at any wavelength from 500 to 700 nm by calibrating the instrument as described below.

Obtaining the Mueller matrix provides at the same time all the polarimetric properties of the sample as its diattenuation (dichroism), retardance (birefringence), and depolarization capability. These quantities could be of major importance for studying pathologies that involve a change of structure. Knowledge of these quantities also permits verification if dichroism is of minor importance compared with birefringence, a hypothesis often made in birefringence measurements.

First, we summarize the essential features of the Stokes–Mueller formalism extensively used in this research. Then, we present the instrument characterization performed with a linear polarizer and a thick wedged quartz plate as test samples. Finally, the imaging capability of our instrument is demonstrated by showing polarimetric images of a stained hepatic biopsy taken between a plate and a cover plate. The biopsy, taken on a patient suffering from liver cirrhosis, presents fibrosis, with a significant amount of collagen. To study the optical properties of the Picrosirius Red stained collagen, we measured the Mueller matrix of this sample at different wavelengths and computed its diattenuation, polarizance, and retardance as a function of wavelength.

2. Stokes–Mueller Formalism

A. Stokes Vector

The Stokes–Mueller formalism[12,13] is used to describe the polarimetric properties of a partially depolarized

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system. Stokes vector $\mathbf{S}$ represents the most general state of polarization of a light wave:

$$\begin{aligned}\mathbf{S}^T &= (I, Q, U, V)\\ &= (\langle I_0 + I_{90}\rangle, \langle I_0 - I_{90}\rangle, \langle I_{+45} - I_{-45}\rangle, \langle I_R - I_L\rangle),\end{aligned} \tag{1}$$

where $I = \langle I_0 + I_{90}\rangle = \langle I_{+45} + I_{-45}\rangle = \langle I_R + I_L\rangle$ is the total intensity, and I_ϕ represents the intensity transmitted by a linear polarizer set at an azimuthal angle ϕ in the plane perpendicular to the propagation direction (with $\phi = 0°, 90°, +45°, -45°$). I_R and I_L are the intensities transmitted by a right or a left circular polarizer. In principle, brackets mean ensemble averages of the intensities. In practice, such ensemble averages are replaced by space and/or time averages. For example, a nonpolarized light will be represented by $\mathbf{S}^T = 1/I(1, 0, 0, 0)$ and a linearly polarized light at an azimuthal angle of 45° by $\mathbf{S}^T = 1/I(1, 0, 1, 0)$. The degree of polarization P corresponding to $\mathbf{S}$ is

$$P = \frac{1}{I}(Q^2 + U^2 + V^2)^{1/2} \le 1. \tag{2}$$

B. Mueller Matrix

Mueller matrix $\mathbf{M}$ of a sample is then defined by the relationship $\mathbf{S}_{out} = \mathbf{M}\mathbf{S}_{in}$, where $\mathbf{S}_{out}$ and $\mathbf{S}_{in}$ are the Stokes vectors of the outgoing light and incoming light; $\mathbf{M} = (m_{ij})$, a 4×4 real matrix with $1 \le i, j \le 4$, is a characteristic of the sample; m_{11} is the sample transmission with unpolarized incident light; m_{01}, m_{02}, and m_{03} describe the diattenuation (dichroism); and m_{10}, m_{20}, and m_{30} characterize the polarizance (capability to polarize an incident depolarized light).

The Mueller matrix of an idealized linear polarizer (transmittance τ) oriented at ϕ is given by

$$\mathbf{M}(\tau, \phi) = \frac{\tau}{2}\begin{pmatrix} 1 & \cos(2\phi) & \sin(2\phi) & 0 \\ \cos(2\phi) & \cos^2(2\phi) & \cos(2\phi)\sin(2\phi) & 0 \\ \sin(2\phi) & \cos(2\phi)\sin(2\phi) & \sin^2(2\phi) & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{3}$$

The Mueller matrix of an idealized dichroic retarder (retardation Δ, partial transmittance ψ) oriented at $\phi = 0°$ is

$$\mathbf{M}(\tau, \psi, \Delta) = \tau\begin{pmatrix} 1 & -\cos 2\psi & 0 & 0 \\ -\cos 2\psi & 1 & 0 & 0 \\ 0 & 0 & \sin 2\psi \cos\Delta & \sin 2\psi \sin\Delta \\ 0 & 0 & -\sin 2\psi \sin\Delta & \sin 2\psi \cos\Delta \end{pmatrix}, \tag{4}$$

where $\Psi = 0°$ or 90° describes a linear polarizer (with its transmission axis at 90° and 0°) and $\Psi = 45°$ corresponds to a retardation plate.

The Mueller $\mathbf{M}(\phi)$ matrix of any sample oriented at an angle ϕ can be deduced from $\mathbf{M}(\phi = 0)$ by the formula $\mathbf{M}(\phi) = \mathbf{U}(\phi)\mathbf{M}(\phi = 0)\ \mathbf{U}(-\phi)$, where the rotation matrix $\mathbf{U}(\phi)$ is defined as

$$\mathbf{U}(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(2\phi) & -\sin(2\phi) & 0 \\ 0 & \sin(2\phi) & \cos(2\phi) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{5}$$

C. Decomposition of a Mueller Matrix

Mueller Matrix $\mathbf{M}$ can be written in terms of polarizance and diattenuation by the following formula[14]:

$$\mathbf{M} = m_{11}\begin{pmatrix} 1 & \mathbf{D}^T \\ \mathbf{P} & \mathbf{m} \end{pmatrix}, \tag{6}$$

where $\mathbf{D}^T = 1/m_{11}(m_{12}\ m_{13}\ m_{14})$, $\mathbf{P}^T = 1/m_{11}(m_{21}\ m_{31}\ m_{41})$, and $\mathbf{m}$ is a 3×3 matrix. $\mathbf{D}$ is the diattenuation vector of $\mathbf{M}$, which is related to the dependence of the sample transmission on the incident light polarization also called dichroism and describes both the magnitude and the axis of diattenuation.

The scalar diattenuation D is given by

$$D = \frac{T_{max} - T_{min}}{T_{max} + T_{min}} = \frac{1}{m_{11}}(m_{12}{}^2 + m_{13}{}^2 + m_{14}{}^2)^{1/2}, \tag{7}$$

where T_{max} and T_{min} are the maximum and minimum transmittances for $\mathbf{M}$. The scalar diattenuation of a linear polarizer and a pure depolarizer are, respectively, 1 and 0. $\mathbf{P}$ is the polarizance vector that describes the sample capability to polarize an incident depolarized light.

Scalar polarizance P is given by

$$P = \frac{1}{m_{11}}(m_{21}{}^2 + m_{31}{}^2 + m_{41}{}^2)^{1/2}. \tag{8}$$

It is easy to demonstrate (using the rotation matrix $\mathbf{U}$) that $P(\phi) = P(\phi')$ and $D(\phi) = D(\phi')$ so that P and D are independent of the sample orientation, an interesting issue in polarimetric images. Any Mueller matrix $\mathbf{M}$ can then be decomposed into a sequence of three matrices: a diattenuator $\mathbf{M}_D$, followed by a retarder $\mathbf{M}_R$, then followed by a depolarizer $\mathbf{M}_\rho$:

$$\mathbf{M} = \mathbf{M}_\rho\mathbf{M}_R\mathbf{M}_D, \tag{9}$$

where

$$\begin{aligned}\mathbf{M}_D &= m_{11}\begin{pmatrix} 1 & \mathbf{D}^T \\ \mathbf{D} & \mathbf{m}_D \end{pmatrix},\\ \mathbf{m}_D &= \sqrt{1 - D^2}\,\mathbf{I} + (1 - \sqrt{1 - D^2})\hat{D}\hat{D}^T,\end{aligned} \tag{10}$$

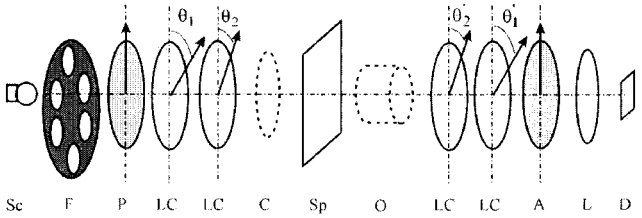


Fig. 1. Schematic of the imaging polarimeter: Sc, white-light source; F, filter wheel; P, linear polarizer (the arrow indicates the direction of the transmission axis); LC, liquid-crystal variable retarders (for each LC, the arrow indicates the direction of its variable index axis); Sp, sample; A, linear analyzer; L, imaging lens (achromat); D, detector (CCD camera). Higher magnifications can be achieved by insertion of a condenser (C) and a microscope objective (O) (dotted lines) into the optical path.

$\mathbf{I}$ is the 3×3 identity matrix, and $\hat{D} = \mathbf{D}/|\mathbf{D}|$.

By calculating the matrix $\mathbf{M}' = \mathbf{M}\mathbf{M}_D^{-1} = \mathbf{M}_\rho\mathbf{M}_R$, we can extract[14] $\mathbf{M}_\rho$ and $\mathbf{M}_R$. The depolarization power (noted as ρ) of the depolarizing Mueller matrix $\mathbf{M}_\rho$ is then given by

$$\rho = 1 - \frac{|Tr(\mathbf{M}_\rho) - 1|}{3}, \qquad 0 \le \rho \le 1. \tag{11}$$

The scalar retardance (birefringence) R is defined between 0 and 180° by

$$R = \cos^{-1}\left[\frac{1}{2}\,Tr(\mathbf{M}_R) - 1\right]. \tag{12}$$

3. Experimental

A. System Setup

Our Mueller imaging polarimeter is outlined in Fig. 1 and is composed of:

A conventional white-light source followed by a wheel of interference filters with center wavelengths of 500, 550, 600, 650, and 700 nm and a 10-nm bandwidth. The spectral range is limited by the CCD camera response for short wavelengths and by the linear polarizer (P) and analyzer (A) for long wavelengths.

The PSG, which generates four linearly independent states of polarization characterized by their Stokes vector $\mathbf{S}_i$. The modulation matrix $\mathbf{W} = (\mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3, \mathbf{S}_4)$ is a 4×4 matrix whose column vectors are the four $\mathbf{S}_i$ Stokes vectors. The PSG is made of one linear polarizer (Melles Griot, 03 FPG 007) and two nematic liquid crystals (PLM20, Jenoptik), LC_1 and LC_2, whose retardations (δ_1, δ_2) depend on their driving voltages. The $\mathbf{W}$ matrix is generated by varying these retardations in a time sequence of the form $(\delta_1, \delta_2) = (\Delta_1, \Delta_1), (\Delta_1, \Delta_2), (\Delta_2, \Delta_1), (\Delta_2, \Delta_2)$.

The sample to be measured, characterized by its Mueller matrix $\mathbf{M}$.

The PSA, which analyzes the polarization of the outgoing light from the sample. Its elements are the same as those of the PSG, but positioned in reverse order. As the liquid crystals are driven by the same sequence as in the PSG, the Stokes vector of the light to be analyzed is projected on four basis states that are the row vectors of the 4×4 analysis matrix $\mathbf{A}$, which characterizes the PSA. The actual values of all the elements of $\mathbf{A}$ and $\mathbf{W}$ are easily determined by the calibration procedure outlined in Subsection 3.B.

A 60-mm achromatic lens that images the sample on a fast silicon CCD camera (Dalsa CA-D1, 256×256 pixels, 12 bits, $16\ \mu m \times 16\ \mu m$ pixels) operated at room temperature. The exposure time was typically set equal to 10 ms. For a sequence we took 16 raw images that yield the 4×4 matrix product $\mathbf{B} = \mathbf{AMW}$ for each pixel. Provided the system is calibrated (i.e., $\mathbf{A}$ and $\mathbf{W}$ are known), matrix $\mathbf{M}$ is readily extracted from raw data $\mathbf{B}$. It is possible to work with higher magnifications by inserting a condenser and a microscope objective just before and just after the sample.

The values of retardations Δ_1, Δ_2 and azimuthal angles θ_1, θ_2 of the liquid crystals (defined with respect to the transmission axis of polarizer P) are optimized as described in Refs. 6, 15, and 16. To minimize the noise in final matrix $\mathbf{M}$ for a given noise in raw data $\mathbf{B}$, the condition numbers of both $\mathbf{A}$ and $\mathbf{W}$ must be minimized. An optimized set of these parameters is[9]

$$\Delta_1 = 315°, \quad \Delta_2 = 135°,$$
$$\theta_1 = \varepsilon 27.4°, \quad \theta_2 = \varepsilon 72.4°, \tag{13}$$

which leads to basis states $\mathbf{S}_i$ that are all elliptically polarized. If this set of parameters is taken for both the PSG and the PSA, then, in principle, $\mathbf{A} = \mathbf{W}^T$. It has been shown[15] that other choices are better suited for particular types of $\mathbf{M}$ matrices, but the choice defined above is an optimal one for the most general case of *a priori* unknown $\mathbf{M}$ matrices.

To obtain results nearly as good as those obtained previously with a He–Ne laser and a single photodiode detector,[9] we had to take into account the nonlinearity of the CCD camera. To this end we measured the CCD response camera as a function of the angle between the transmission axes of two linear polarizers, and we used the well-known Malus law to define the linear scale of illumination. This CCD nonlinearity has been found to depend only on the output signal level, and it is approximately 10% when this signal is close to saturation.

Great care has been taken to get rid of stray light. The offset that is due to both residual stray light and to dark current is determined and subtracted from the measured signal for each pixel. To reach a satisfactory signal-to-noise ratio we average typically 15 images for each state of polarization. At this level of averaging, random errors become comparable to residual systematic errors. Since the liquid-crystal switching time is 60 ms, a complete Mueller matrix image is taken in only 5 s.

B. Calibration

We briefly describe the calibration method of the instrument; for more details see Ref. 17. An essential

feature of this method is that it provides the actual matrices **A** and **W** without modeling the operation of the polarimeter. For this reason, no precise mechanical mounts are needed for elements of the PSG and PSA, and many effects, such as the fact that the degree of polarization $P(\mathbf{S}_i)$ of the basis states is degraded when the instrument is illuminated with a relatively broad spectrum, are automatically taken into account. The essential steps are the following:

First, no sample is inserted between the PSG and the PSA. The instrument measures the matrix $\mathbf{B}_0 = \mathbf{AW}$.

Then, a reference sample is put into the beam. The instrument measures the matrix $\mathbf{B} = \mathbf{AMW}$, where **M** is the Mueller matrix of the reference sample. If we make the product $\mathbf{C} = \mathbf{B}_0^{-1}\mathbf{B} = \mathbf{W}^{-1}\mathbf{MW}$, we obtain matrix **C**, which has the same eigenvalues as **M**. Provided that the sample is a dichroic retarder characterized by three parameters (τ, Ψ, Δ), its Mueller matrix is entirely determined by its (experimentally known) eigenvalues, except for a rotation angle ϕ related to the azimuth of the sample axes. In practice we use a dichroic polarizer set at $\phi_1 = 0°$ by definition and at approximately $\phi_2 = 90°$ and a retardation plate (with a retardation value anywhere from 80° to 140°) set at approximately $\phi_3 = 30°$.

Matrix **W** (together with the azimuths of the reference samples) is determined by the set of commutation equations $\mathbf{C} = \mathbf{W}^{-1}\mathbf{MW}$, which must be verified for all the reference samples.

Finally, matrix **A** is calculated by the formula $\mathbf{A} = \mathbf{B}_0\mathbf{W}^{-1}$.

With our camera, a complete pixel by pixel calibration would be too long. So, we chose to calibrate the instrument on 225 pixels of coordinates (X_i, Y_i), spread over the detector surface yielding the $\mathbf{W}(X_i, Y_i)$, and then we calculated an interpolated $\langle\mathbf{W}\rangle$ matrix for a pixel of coordinate (X, Y) with the formula

$$\langle\mathbf{W}(X, Y)\rangle = \frac{\sum_{i=1}^{225} f_i(X, Y)\mathbf{W}(X_i, Y_i)}{\sum_{i=1}^{225} f_i(X, Y)},$$

where

$$f_i(X, Y) = \exp(-a|(X - X_i)^2 + (Y - Y_i)^2|). \quad (14)$$

The a factor was chosen so that weighting functions f_i can significantly overlap. This type of interpolation has the advantage of creating no sharp artifacts in the polarimetric images. The interpolated $\langle\mathbf{A}\rangle$ matrix is then calculated for each pixel by the formula $\langle\mathbf{A}\rangle = B_0^{-1}(X, Y)\langle\mathbf{W}(X, Y)\rangle$. The calibration procedure takes a few minutes. We regulated the temperature of the liquid crystals at 35 °C ± 0.1 °C, and the instrument stability is such that calibration is not needed more than once per day.

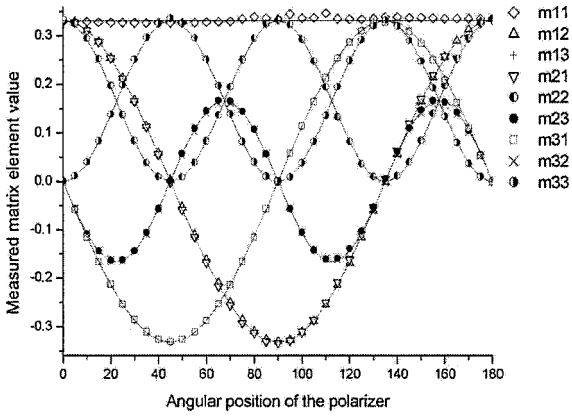


Fig. 2. Mueller matrices measured with a linear polarizer set at various azimuthal angles. Only the nonvanishing elements (symbols) and single parameter fits (solid curves) have been represented.

4. Evaluation of Instrument Accuracy

A. Linear Polarizer

To evaluate the overall system performance we took a series of Mueller matrix measurements with a dichroic sheet polarizer (Melles Griot FPG 007) measured at different azimuthal angles. The measured Mueller matrices appear to be uniform overall the images (typical standard deviation of 5×10^{-4}). A typical result at 650 nm is shown in Figs. 2 and 3 for one pixel. The absolute value of the angular position of the polarizer was known with the accuracy of the angle reading (≈0.3°). The elements of the 3×3 upper diagonal block, expected to follow Eq. (3), are drawn in Fig. 2. All the data have been fitted with a single parameter, the polarizer transmission, with an optimal value of 0.662. We note a small fluctuation of element m_{11} (variation from 0.32 to 0.34), a fact that could be due to an intensity fluctuation of the light source. Figure 3 shows that the other elements, expected to vanish, are actually quite small (absolute value lower than 6×10^{-3}).

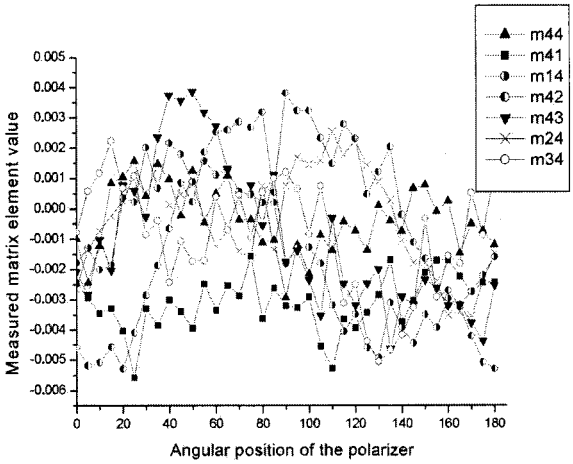


Fig. 3. Same experiment as in Fig. 2 with elements that are expected to vanish.

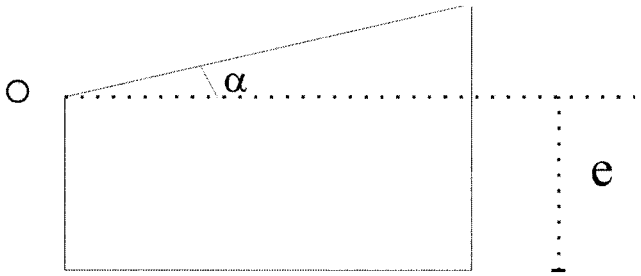


Fig. 4. Schematic of the wedged quartz plate. The optical axis is parallel to the x axis.

Note that such accuracy is probably the best ever reported for Mueller imaging polarimeters, and that it is readily obtained for a variety of conditions (variable wavelengths and magnifications), because of the robustness of the calibration procedure.

B. Wedged Quartz Plate

As an alternative test sample, we used a wedged quartz plate with its optical axes parallel to the wedge direction and sufficiently thick to exhibit a significant depolarization when illuminated with a 10-nm wide spectrum (Fig. 4). For such a sample, a description in terms of a nondepolarizing dichroic retarder Eq. (4) is not possible. As a consequence, Mueller ellipsometry instead of classical ellipsometry must be used. For this plate and for a given wavelength λ, retardation can be written as

$$\Delta(x) = \frac{2\pi\delta}{\lambda}\, x\, \tan(\alpha) + \frac{2\pi\delta}{\lambda}\, e, \tag{15}$$

where $\delta = 9.04 \times 10^{-3}$ at $\lambda = 650$ nm, $\alpha = 3° 00'$, and $e \cong 3$ mm. Retardance [defined by Eq. (12)] is expected to reach 180° on two pixels separated by

$$i = \frac{\lambda}{\delta\, \tan(\alpha)} \cong 1.37 \text{ mm}. \tag{16}$$

Figure 5 represents a retardance image of the plate that was positioned in such a way that one of its index axes is parallel to the lines of the CCD camera. The dashed line indicates which row was chosen for curves 6 and 7. As expected, the measured retardance variation is linear between 0 and 180° (Fig. 6). Retardance reaches its maximum for two pixels separated by (108 ± 2) pixels corresponding to (1.36 ± 0.03) mm, which is in good agreement with the expected value of 1.37 mm.

If we plot the depolarization power as a function of pixel number in Fig. 7, the depolarization power decreases from left to right, as the plate becomes thinner. There are two local maxima localized where the retardance is close to 0°. Indeed, if we state that measurement $\mathbf{M}$ is the sum between two retardations

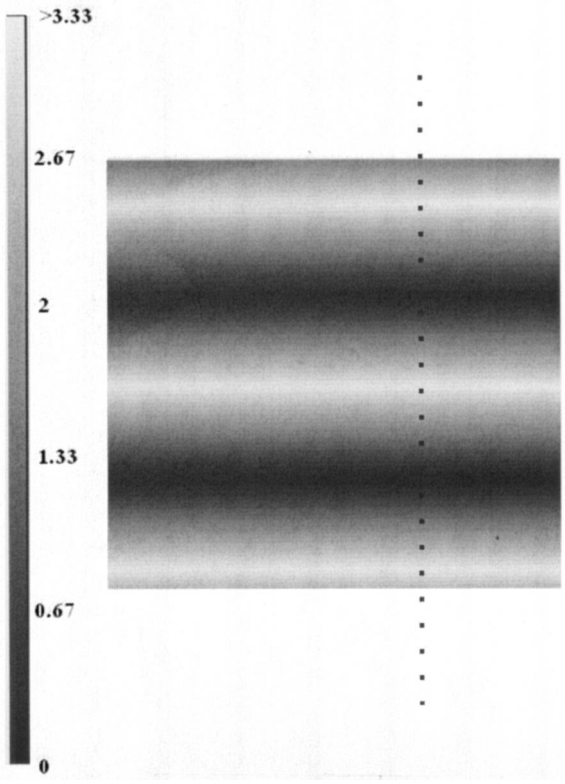


Fig. 5. Retardance (rad) image of the wedged quartz plate. The field of view is 2.7 mm × 2.7 mm.

Δ_1 and Δ_2 of the Mueller matrix of a dichroic retarder [Eq. (4), taking $\psi = 45°$], then

$$\mathbf{M} = \frac{1}{\Delta_2 - \Delta_1} \int_{\Delta_1}^{\Delta_2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\Delta & \sin\Delta \\ 0 & 0 & -\sin\Delta & \cos\Delta \end{bmatrix} \mathrm{d}\Delta$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & a = (\sin\Delta_2 - \sin\Delta_1)/(\Delta_2 - \Delta_1) & -b \\ 0 & 0 & b = (\cos\Delta_2 - \cos\Delta_1)/(\Delta_2 - \Delta_1) & a \end{bmatrix}. \tag{17}$$

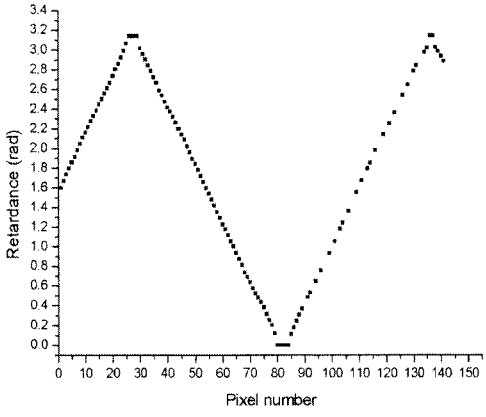


Fig. 6. Retardance as a function of pixel number for the row indicated in Fig. 5.

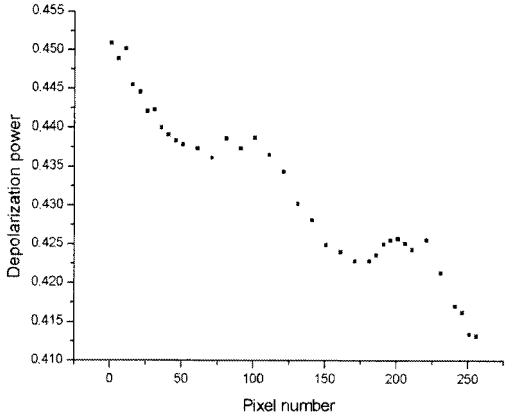


Fig. 7. Depolarization power as a function of pixel number for the row indicated in Fig. 5.

It is easy to demonstrate, with a polar decomposition, that with a fixed value of the integration range $\Delta_2 - \Delta_1$, the depolarization power of this matrix is higher for an average retardation near 0°. In the first approximation, the light wavelength is included between $\lambda_1 = 655$ nm and $\lambda_2 = 645$ nm and the width of the plate is equal to $e \cong 3$ mm. Then, $\Delta_1 \approx (2\pi\delta e/\lambda_1) \cong 260.15$ rad and $\Delta_2 \approx 2\pi\delta e/\lambda_2 \cong 264.19$ rad. As a result, depolarization in this case is mostly due to the light bandwidth (10 nm). Using Eq. (17), the measured matrix can be estimated as

$$\mathbf{M} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -0.125 & -0.28 \\ 0 & 0 & 0.28 & -0.125 \end{pmatrix},$$

which gives, after polar decomposition, a depolarization power of 0.46, close to the average measured depolarization power.

5. Polarimetric Images of a Hepatic Biopsy

A. Background

We chose to study hepatic biopsies taken on patients suffering from liver cirrhosis, a well-known disease, which involves a modification of hepatic tissue with deposition of collagen. The diagnosis of cirrhosis is established by a pathologist on a percutaneous liver biopsy. To the exception of the limited area around vessels, there is no collagen in normal liver. A four-stage grading system (F1–F4) has been developed based on fibrosis extent, i.e. on the fraction of normal liver parenchyma replaced by collagen. Grade F1 is benign; grade F4 is the most aggressive. In early stages, fibrosis is confined to areas close to the vessels and extends in latter stages to the whole liver. In F4 stage cirrhosis, the normal tissue architecture has disappeared and fibrotic tracts isolate liver nodules. The correlation between the stage of the disease and the clinical consequences of cirrhosis is clear but not yet really satisfactory. Because of the anisotropy of collagen fibers, polarimetric imaging is likely to be particularly well suited for an accurate characterization of this type of tissue modification. This technique might eventually provide a finer grading scale aimed at improving therapeutic strategies. We chose to work with a hepatic biopsy, approximately 0.5 mm × 10 mm in size and 10 μm thick, taken on one patient, at stage F4. The sample was stained with a specific collagen dye (Picrosirius Red). Fibrosis then appears purple, hepatocytes appear pink. Figure 8 shows a high magnification image of a stained sample, where the difference between the structures of normal and fibrotic tissues is obvious, the latter appearing darker.

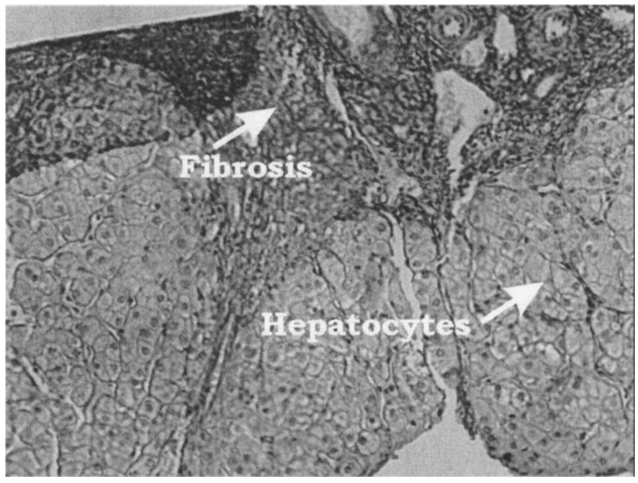


Fig. 8. Typical hepatic biopsy stained with Picrosirius Red (classical nonpolarized microscopic image). The field of view is 0.5 mm × 0.7 mm. Fibrotic zones appear darker than the normal tissue that consists of clearly visible hepatocytes. Courtesy of Thierry Boulesteix, E. Beaurepaire, and Marie Claire Schanne-Klein, Laboratoire d'Optique et Biosciences, Centre National de la Recherche Scientifique, Ecole Polytechnique, France.

Picrosirius Red dye has long been known to enhance contrast for samples with collagen fibers observed with a polarizing microscope. This enhancement is certainly due to the fact that Picrosirius Red molecules orient themselves parallel to collagen fibers: such a microscopic anisotropy might induce local birefringence or diattenuation (dichroism), or both. Typically, circularly rather than linearly polarized light must be used to obtain a contrast independent of the sample orientation.[18]

B. Polarimetric Images and Determination of Optical Constants of Stained Collagen

In the following, the sample (grade F4) taken between a plate and a cover plate is imaged and measured at different wavelengths. The unpolarized transmission image m_{11} is shown in Fig. 9 for three wavelengths: 500, 550, and 650 nm. The transmission image is not contrasted at 650 nm compared with 500 and 550 nm because at this wavelength the dye does not absorb the incoming light. Diattenuation, a quantity independent of the sample orientation, has been calculated and represented pixel by pixel in Fig. 10. Fibrotic zones that isolate the liver in nodules appear white,

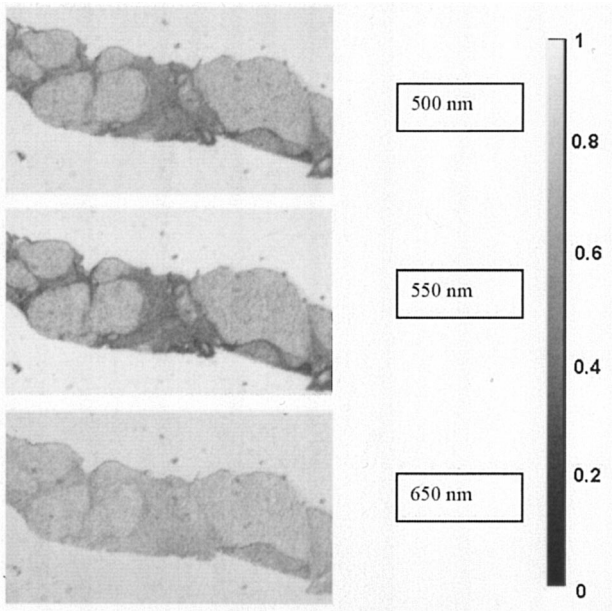


Fig. 9. Unpolarized image (m_{11}) of a stained hepatic biopsy taken between a plate and a cover plate for three different wavelengths. The patient suffers from liver cirrhosis, which separates the liver into nodules. The field of view is 2 mm × 2 mm.

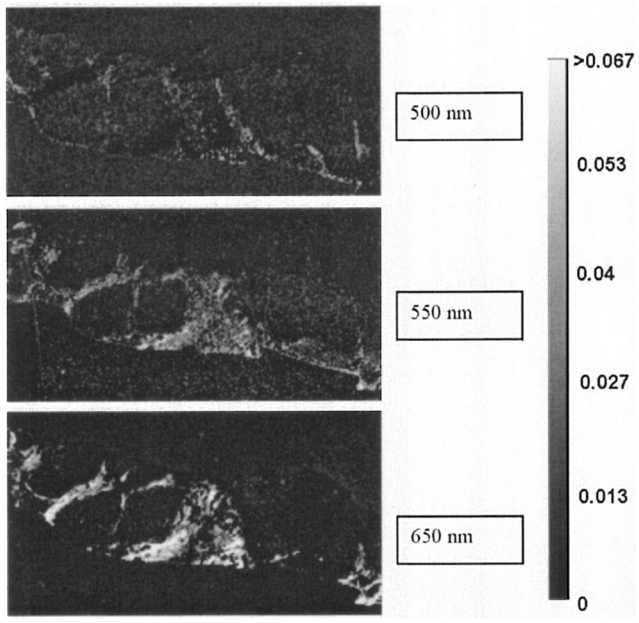


Fig. 11. Retardance (rad) image of the same sample as in Fig. 10. The field of view is 2 mm × 2 mm.

providing a cartography of the fibrosis concentration in the sample. The diattenuation signal is significantly higher at 550 nm than for the other two wavelengths.

Retardance has been calculated and represented pixel by pixel in Fig. 11. This time retardance is higher at 650 nm than for other wavelengths. The image is a bit different from the one obtained in diattenuation, suggesting that the physical properties contained in dichroism and birefringence are not identical. The birefringence maximum is approximately 0.1 rad for 650 nm. If we suppose that the fibrosis thickness is approximately 10 μm for this maximum, we can estimate that the birefringence is approximately $\delta \approx 10^{-3}$ at 650 nm, which is in agreement with the reported value of collagen birefringence.[19] However, this birefringence is not due to intrinsic collagen birefringence but to the system formed by the dye and the collagen. Depolarization power is shown in Fig. 12. At 550 nm, depolarization power is highest (but stays low). This depolarization power does not seem to correlate directly with

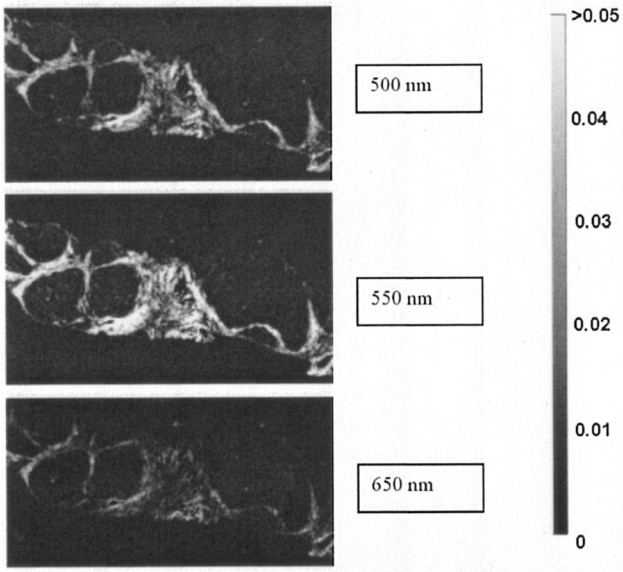


Fig. 10. Diattenuation image of the same sample as in Fig. 10. The field of view is 2 mm × 2 mm.

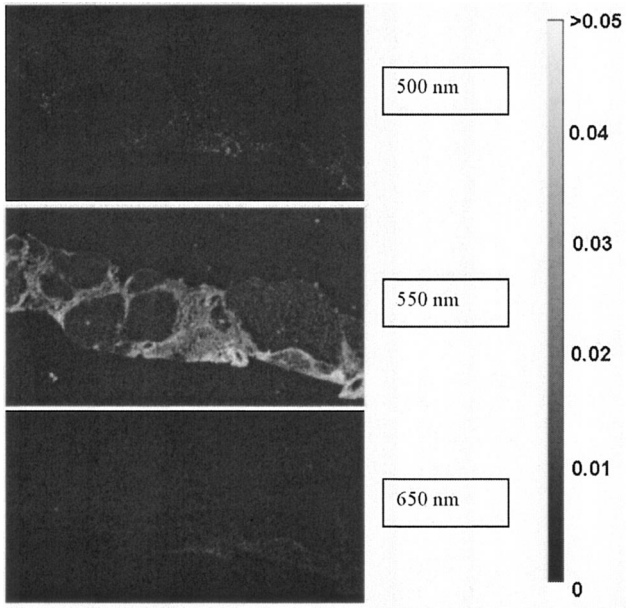


Fig. 12. Depolarization power image of the same sample as in Fig. 10. The field of view is 2 mm × 2 mm.

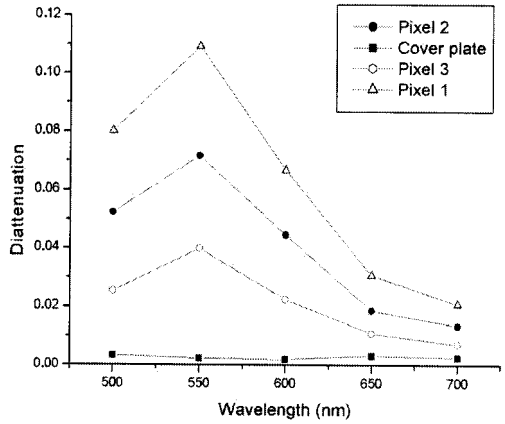


Fig. 13. Diattenuation as a function of wavelength for three different pixels chosen for fibrosis and for a point of the cover plate.

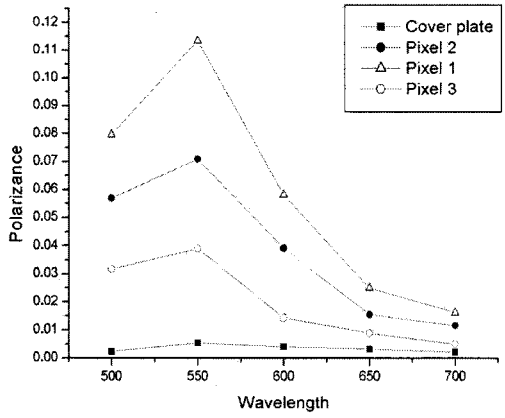


Fig. 14. Polarizance as a function of wavelength for three different pixels chosen for fibrosis and for a point of the cover plate.

the absorption of light because at 500 nm the depolarization power is low in spite of high absorption.

We have represented the diattenuation, polarizance, and retardance as a function of wavelength for three pixels situated on fibrosis and for another one situated on the cover plate (Figs. 13, 14, and 15).

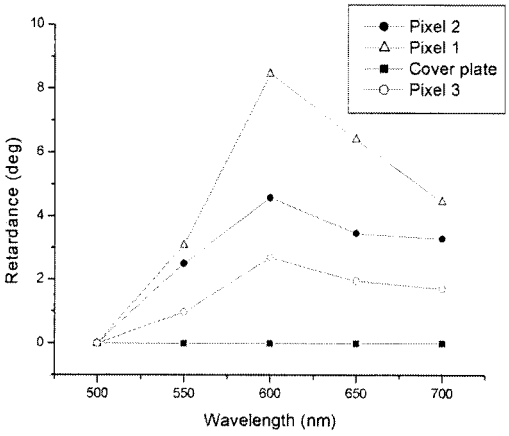


Fig. 15. Retardance as a function of wavelength for three different pixels chosen for fibrosis and for a point of the cover plate.

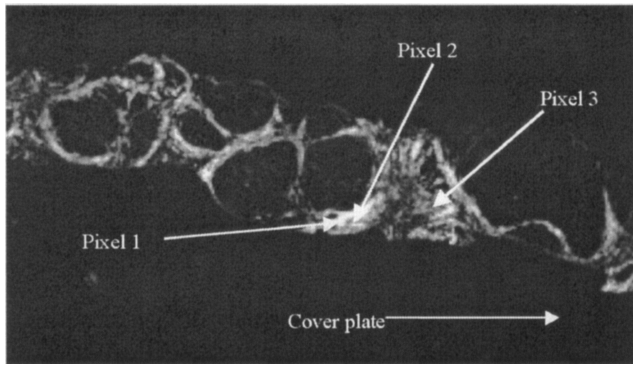


Fig. 16. Localization of the four pixels taken into account in Figs. 14–16 for a diattenuation image of the sample at 550 nm. The field of view is 3 mm × 2 mm.

The expected values of these three quantities for the cover plate should be ideally equal to zero, giving an estimation of error measurements. The four pixels taken on account are shown in Fig. 16. In spite of the low sensitivity of the CCD camera for wavelengths less than 600 nm, the measurement accuracy remains satisfactory. Indeed, diattenuation and polarizance of the cover plate remains lower than 5×10^{-3}. Note that healthy tissue also has a low polarimetric response.

Both the diattenuation and the polarizance reach a maximum at 550 nm. The value of this maximum depends on the local concentration of collagen but the spectral behavior of both quantities is always the same for a point situated on a fibrotic zone. At this wavelength, these quantities are approximately five times higher than at 700 nm. Retardance reaches its maximum at 600 nm and becomes low at 500 nm. These results suggest that the depolarization seen at 550 nm could be due to diattenuation rather than to retardance inhomogeneities in the tissue. Finally, the contrast seen with a polarizing microscope operating at 500 nm is mostly due to dichroism, although it is due to both dichroism and birefringence for other wavelengths.

6. Conclusion

We have demonstrated a Mueller polarimetric imaging system based on liquid-crystal modulators and a spectrally filtered white-light source. The instrument performance is satisfactory with an average 5-s data acquisition and good accuracy. As an example of a high depolarizing sample, we studied a thick wedged quartz plate. Polarimetric images of a stained hepatic biopsy taken at different wavelengths on a patient suffering from liver cirrhosis have been shown. These images have been used to plot diattenuation, polarizance, and retardance as a function of wavelength for a detailed investigation of the optical properties of the Picrosirius Red stained collagen.

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